

PORT AUGUSTA

GEOLOGICAL SURVEY OF SOUTH AUSTRALIA
DEPARTMENT OF MINES ADELAIDE

S.A. GEOLOGICAL ATLAS SERIES SHEET SI 534 ZONE 5

FIRST EDITION 1968



REFERENCE

- Qb** White sand of the coastal dunes (SEMAPHORE SAND), modern beach sand and gravel.
- Qrs** Sand of the gypsum dunes: Aeolian beds mapped with YAMBA FORMATION.
- Qra** Fluvial sands and gravels of the modern drainage channels.
- Qrs** Sand sheets and dune sands (red-brown FULHAM SAND and pale-yellow MOULNAUX SAND equivalents).
- Qr** ST. KILDA FORMATION: Sands, shelly silts and clays of the littoral lagoons and mangrove swamps.
- Qrl** YAMBA FORMATION: Lacustrine beds.
- Qrl** Flour gypsum of the sample flats.
- Qrn** Grey plastic gypsiferous muds.
- Qrt** Soil mantles, including gypsiferous LOVEDAY SOIL with a characteristic soft platy and nodular carbonate horizon.
- Qpk** POORAKA FORMATION: Gravels, sands and clayey sands with clay lenses. Overlies lower Pleistocene HINDMARSH CLAY in many places.
- Qpt** Calcrete in BAKARA SOIL with RIPON CALCRETE at the base.
- Tfe** TELFORD GRAVEL: Dissected high-level gravels, calcreted in places.
- Ts** Laterite as cappings and exposed on low rises.
- Ts** Silcrete cappings, thin boulder beds and sandstones containing leaf impressions.

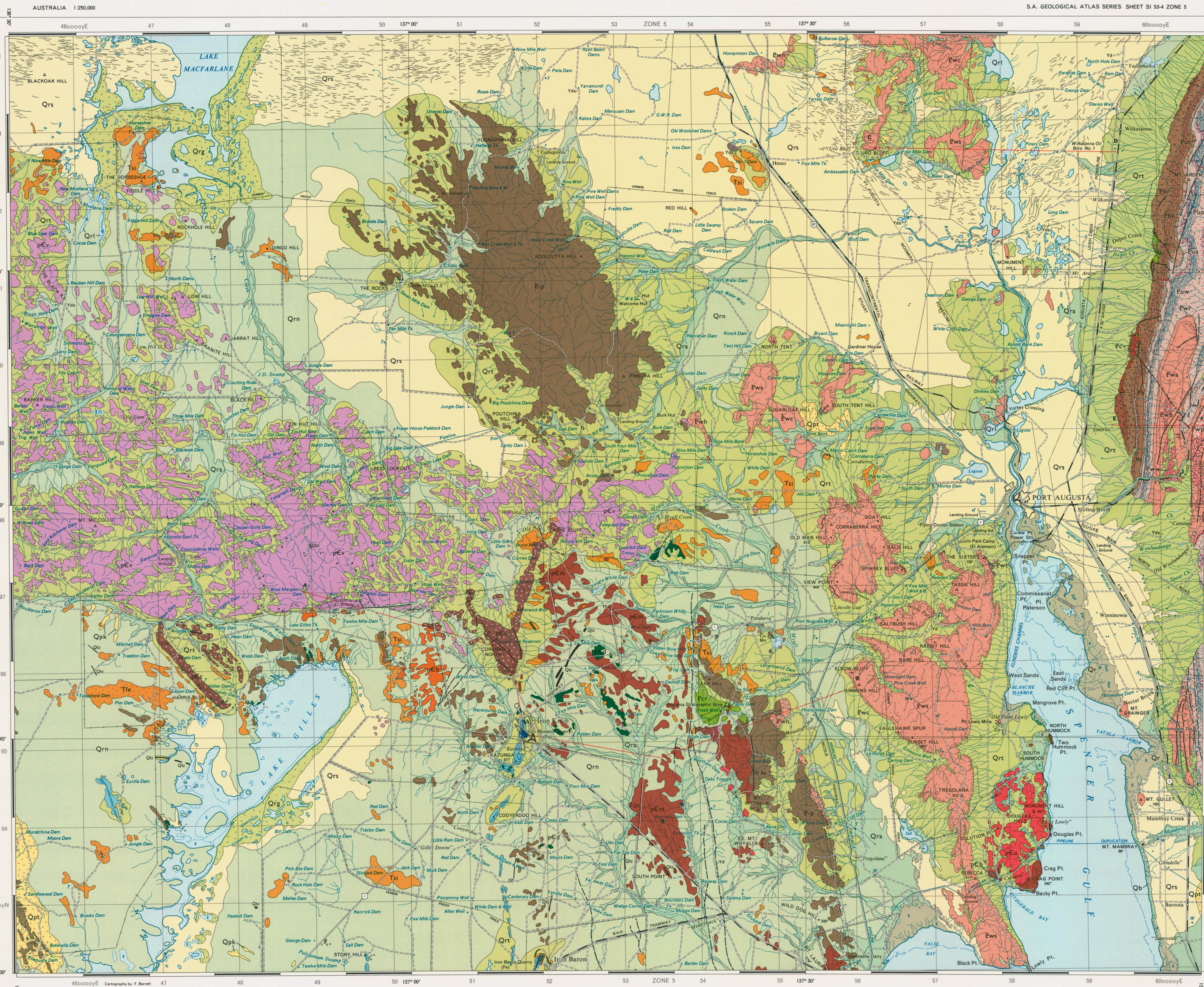
SEQUENCE IN THE ADELAIDE GEOSYNCLINE EAST OF PORT AUGUSTA

- Pwp** POUND QUARTZITE: White quartzite and underlying red sandstones of Devils Peak. Top not exposed.
- Pw** WONOKA FORMATION: Grey flaggy calcareous siltstones.
- Pw** SUNNYBROOK FORMATION: Purple and grey-green shales.
- Pw** A.B.C. RANGE QUARTZITE: Thin bedded, ripple marked quartzite with silty bands and heavy mineral laminae.
- Pw** BRACHINA FORMATION: Red and green shales, siltstones.
- Pw** NUCCAULENA FORMATION: Lenticular pink dolomite.
- Pw** WILLOCHRA FORMATION: Upper unit: Rippled sandstones, siltstones, gritty limestone. Lower unit: Red-grey calcareous siltstones, mud cracked.
- Pw** BRIGHTON Limestone EQUIVALENT: Grey massive dolomite.
- Pw** TAPLEY HILL FORMATION: Green calcareous siltstone.
- Pw** TINDLEPINE SHALE MEMBER: Finely laminated dolomitic shale.
- Pw** APPILA TILLITE EQUIVALENT: Boulder beds, quartzite, siltstone.
- Pw** Unnamed dolomitic shales and quartzite beds.
- Pw** SKILLOGALEE DOLOMITE: Laminated dark grey and pale dolomite, chert grey siltstone, bedded quartzite; laminated magnesian beds, magnesian conglomerate.
- Pw** RHINE SANDSTONE: Pink and pale grey feldspathic quartzite, laminated and flaggy; minor pink and grey siltstone pebbly in places near base; ripple marks, cross-bedding, heavy mineral laminae, mud cracks. (EMERO QUARTZITE, includes NELSHAY SANDSTONE, WIRABARA FORMATION).
- Pw** Quartzite and siltstone: flaggy laminated, pinkish grey and greyish-red; ripple marks, cross-bedding, heavy mineral laminae and salt cracks.
- Pw** Conglomerate, pebbly quartzite; rounded pinkish quartzite pebbles in silty matrix, some trachyte pebbles.
- Pw** Heavy mineral laminae, cross-bedding.
- Pw** ROOPENA VOLCANICS EQUIVALENT at Depot Creek: Purple and green amygdaloidal trachyte, minor siltstone and pyroclastic tuffites. Thin grit at base with trachyte pebbles.
- Pw** Reddish siltstone, sandstone, dolomite, silty dolomite; laminated, ripple marks, mud cracks.
- Pw** Diapiric breccia with carbonate matrix; rafts of contorted siltstones, sandstones and dolomites.

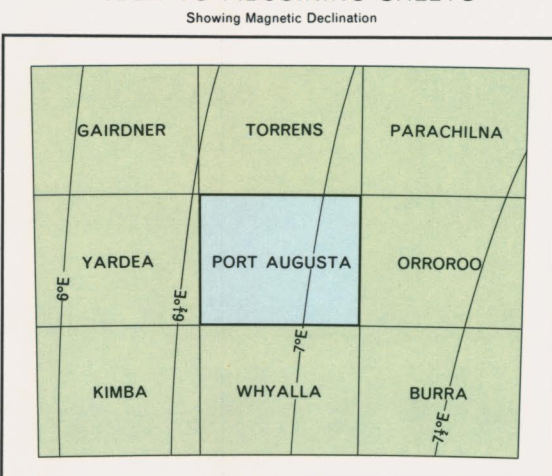
UNITS OF THE STUART SHELF AND GAWLER PLATFORM

- Eh** (Section only) PARARA, KULPARA LIMESTONES and PARACHILNA FORMATION EQUIVALENTS from the Wilkesia Series.
- Eh** Unconformity.
- Eh** TERT HILL FORMATION: Equivalent of A.B.C. RANGE QUARTZITE and BRACHINA FORMATION.
- Eh** SIMONS QUARTZITE MEMBER: Dense white quartzite.
- Eh** CORRAHERA SANDSTONE MEMBER: Red-brown micaceous sandstone.
- Eh** TREGOLANA SHALE MEMBER: Purple and brown sandy siltstones and shales. Equivalent of WOOLMERA SHALE.
- Eh** WHALLA SANDSTONE MEMBER: Poorly outcropping basal grits and sandstones with abundant cobbles of Pandurra Formation.
- Eh** Unconformity.
- Eh** PANDURRA FORMATION: Medium-grained and gritty red sandstones, cross-bedded, conglomeratic near base.
- Eh** Unconformity.
- Eh** ROOPENA VOLCANICS: Purple and grey-green amygdaloidal trachytes, basalt and andesite with minor diorite and also altered amygdaloidal flow rocks of Two Hummock and Backy Point. Equivalent of volcanic of Depot Creek.
- Eh** BACKY POINT FORMATION: Thin bedded sandstone with cobbles, unconformable on MOONABE FORMATION.
- Eh** Unconformity.
- Eh** CULTANA GRANITIC COMPLEX: Granite with porphyritic and granoblastic textures. Intrudes MOONABE FORMATION. Overlain by BACKY POINT FORMATION in Cultana area.
- Eh** Unconformity.
- Eh** GAWLER RANGE VOLCANICS: Quartz-feldspar porphyry, columnar jointed rhyolite, rhyodacite, dacite and welded vitro and lithic tuffs. Intruding into and overlying CORUNNA CONGLOMERATE.
- Eh** CORUNNA CONGLOMERATE: Conglomerate, sandstone, brown and dark grey siltstone, dolomite, crystalline.
- Eh** Unconformity.
- Eh** BURKITT GRANITE: Massive hornblende granite. Overlain by CORUNNA CONGLOMERATE.
- Eh** Basic intrusive. Amphibolite, serpentinite, dolerite, gabbro in sills, dykes and plugs intruding OLDER PRE-CAMBRIAN and MOONABE FORMATION.
- Eh** MOONABE FORMATION: Indurated dark-grey quartzite with scattered rhyolite pebbles. Commonly a granular greywacke. Gently folded.
- Eh** Inferred regional unconformity.
- Eh** Bands and bodies of granite and pegmatite associated with gneisses (including FLINDERS GNEISS), developed in Lower Proterozoic sediments during early Carpentarian metamorphic event.
- Eh** WARROW QUARTZITE: Coarse recrystallized glassy quartzite and quartz-schist schists.
- Eh** MIDDLEBACK GROUP: Jaspilites (associated with iron ore bodies in Middleback Ranges).
- Eh** Unidentified schist, dolomite, chert and amphibolites (including COOKS GAP SCHIST and MIDDLEBACK NORTH DOLOMITE).
- Eh** Lower Proterozoic rock units, largely converted to early Carpentarian gneisses and now represented by local relic schist and amphibolite bodies (not differentiated on map).

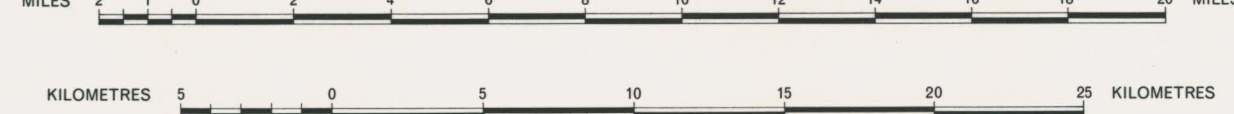
- GEOLOGICAL BOUNDARIES.**
- FAULTS.**
- FOLDS.**
- PLUNGE OF DRAG FOLD.**
- BEDDING.**
- VERTICAL.**
- HORIZONTAL.**
- OVERTURNED.**
- TREND OF BEDDING.**
- TOP OF BED. CROSS-BEDDING.**
- FOLIATION INCLUDED.**
- FOLIATION VERTICAL.**
- FOLIATION WITH PLUNGE OF LINEATION.**
- JOINTING.**
- INCLUDED.**
- VERTICAL.**
- FLOW BANDING.**
- TYPE SECTION.**
- PLANT FOSSIL LOCALITY.**
- MAIN ROAD.**
- SECONDARY ROAD.**
- TRACK.**
- NATIONAL ROUTE NUMBER.**
- RAILWAY.**
- BOUNDARY FENCE.**
- TRIANGULATION STATION.**
- EPHEMERAL STREAM.**
- LAKE.**
- SWAMP.**
- MANGROVES.**
- WATER FEATURES.**
- BORE.**
- WATER HOLE.**
- WELL.**
- EARTH TANK OR DAM.**
- QUARRY.**
- MINE.**
- BARITIES.**
- IRON.**
- QUARTZ.**
- COPPER.**
- MANGANESE.**
- HEAVY METAL ORE (HIGH GRADE).**
- PORT AUGUSTA SHEET SI 534**



INDEX TO ADJOINING SHEETS



SCALE

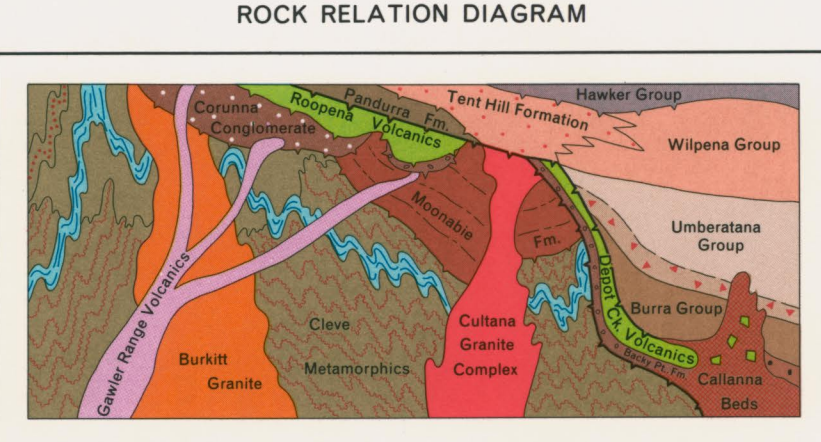


H. J. WALL GOVERNMENT PHOTOGRAPHER, ADELAIDE

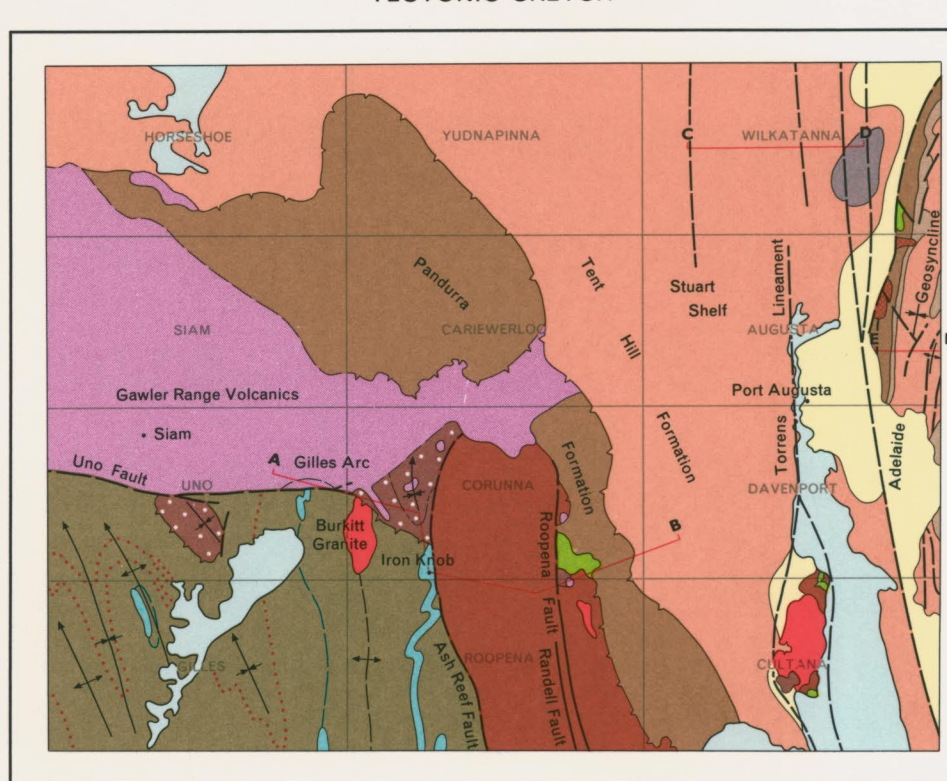
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ROCK RELATION DIAGRAM



TECTONIC SKETCH



- Quaternary**
- Cambrian**
- Adelaidian**
- Marinoan**
- Sturtian**
- Torrensian-Willouran**
- Willouran**
- Carpentarian**
- Lower Proterozoic**
- Vertical**
- Flow banding**
- Plant fossil locality**
- Main road**
- Secondary road**
- Track**
- National route number**
- Railway**
- Boundary fence**
- Triangulation station**
- EpheMERAL stream**
- Lake**
- Swamp**
- Mangroves**
- Water features**
- Bore**
- Water hole**
- Well**
- Earth tank or dam**
- Quarry**
- Mine**
- Barities**
- Iron**
- Quartz**
- Copper**
- Manganese**
- Heavy metal ore (high grade)**



Geological compilation and revision by C. R. Dalgle, M.Sc., J. E. Johnson, B.Sc., G. Forbes, Ph.D., & P. Thomson, M.Sc., from previously published Atlas Series Maps and unpublished Geological Survey regional mapping. Supervising Geologist, Regional Survey Division, Department of Mines, S.A.

Map preparation by Cartographic Section, Department of Mines, S.A.

Compiled under the direction of T. A. Barnes, Government Geologist, Director of Mines.

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